KU LEUVEN

Lab seminar

Doing research with AI agents

Rigour made affordable: a knowledge tree of inquiries

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Damien Bonnet-Eymard

FLANDERS MAKE
DRIVING INNOVATION IN MANUFACTURING

RELIABLE & ROBUST DESIGN

What happens when an agent does research unchecked?

! It cites what it never read

Plausible references and values that do not exist.

Walters & Wilder 2023

! It tries forty variants, reports one

The garden of forking paths, now nearly free.

Gelman & Loken 2014

! It agrees with you

Models tend to tell users what they want to hear.

Sharma et al. 2024

Research has long known the cure: preregistration, verification, published negatives, review. It could rarely afford all of it.

Agents make rigour affordable: twelve rules, four groups

1 Planning

Fix the protocol before the runs: explore or confirm.

fixes: *testing what the data suggested*

2 Evidence

Every number traces to a run, a derivation or a page.

fixes: *numbers nobody can rebuild*

3 Claims

Publish the negatives and the denominator; cite the page you checked.

fixes: *forty variants, one reported; invented citations*

4 The tree

The human confirms, after a blind review.

fixes: *the agent agreeing with you*

12 rules, 5 of them fixed. The agent shows them to you at the start, and asks before you set one aside.

Grounded in the reproducibility literature: Ioannidis 2005; Munafò et al. 2017; full map in docs/literature.md.

Anatomy of an agent: own the top layer

Your framework
the memory

Plain Markdown in your folder: AGENTS.md · .agents/skills/ · docs/wiki/ · inquiry.yaml

Harness
the hands

Reads and writes files, runs code, loops until done: Claude Code · Codex · Gemini CLI · OpenCode

LLM
the brain

Predicts the next token; knows nothing of your project: Claude · GPT · Gemini · Qwen

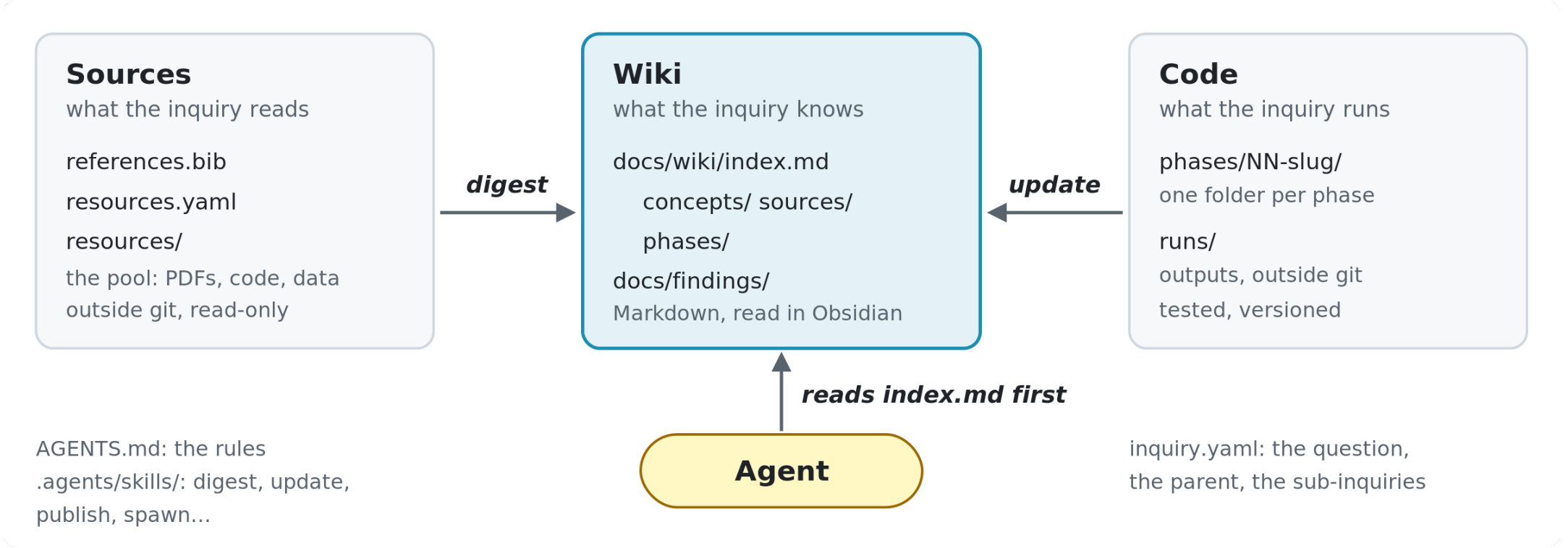
Own the top layer

Models and harnesses change every few months, and so do the leaders.

Keep what you build in plain Markdown in your folder: switch model or harness, keep your knowledge.

AGENTS.md is read by most harnesses; Claude Code reads it through CLAUDE.md.

One folder, one question: sources, wiki, code



Sources are digested into the wiki, code changes update it, and the agent reads index.md first and cites the notes it used.

The wiki has two readers

You, in Obsidian

Browse from index.md, follow the links

The graph: how sources and concepts connect

“What did we take from X?” in seconds

The agent, as context

Reads index.md before answering

Lands on the 2–3 right notes, not 40 PDFs

Cites the notes, so you can check

illustrative

```
# Kooijman et al. (2011): a bicycle stable
#   without gyroscopic or caster effects

[@kooijman2011bicycle]. A purpose-built bicycle
with neither effect still balances itself.

## What we take
- **Neither effect is necessary.**
  → [gyroscopic](../concepts/gyroscopic.md)

## What we don't take
Their prototype's mass layout, for now.
```

One note per source or concept. A digest is a selection, not a summary: what the inquiry uses or rejects, with a link to where.

When a question is too broad, it grows a tree

```
bicycle/              why does a moving bicycle
│                     stay upright?
└── inquiries/
    ├── benchmark-model/
    │   └── inquiries/speed-sweep/
    ├── gyroscopic/
    └── trail/
```

Zoom in

“Spawn a sub-inquiry for the gyroscopic effect”: a child repository with its own question, under its parent.

Zoom out

Adopt an existing repository under a broader question; merging two is a new root and two adopts.

Knowledge flows

Down by link, up by findings, one level at a time; never sideways.

Every index routes to its children, the open questions and what is out of scope: asked a broad question, the agent descends one index at a time.

What it looks like

docs/wiki/index.md, at the root

```
## Sub-inquiries
- [gyroscopic](…/index.md): how much does
  wheel spin contribute? Active.
  Dive here for: wheel spin, precession.
- [trail](…/index.md): …

### Open
- How much does the rider's steering
  contribute? (no inquiry yet)
```

docs/findings/…, at the root

illustrative

```
status: hint      # its children are hints
tried: 3          # explanations tested
evidence:
  - kind: sub-inquiry
    finding: …/gyroscopic/blob/8be41d0/…
    status: hint

# Neither wheel spin nor trail alone
# keeps the benchmark bicycle upright
Scope: the model only; not yet validated.
```

rkt check verifies what a machine can tell: links, routes, the denominator, no parent stronger than its children.

Limits

! Errors get rarer, not impossible

The rules and the checks make invention harder.

Check what matters against the source.

! Confidentiality

Files and prompts go to the model provider.

Keep confidential material out of the inquiry; deny it in your harness's settings.

! Cost and lock-in

Subscriptions, quotas and leaders change.

Plain Markdown keeps your work portable.

You remain the author: of the questions, of the code you merge, and of every claim you sign.

The tools will change. Who decides should not.

Models and harnesses improve every few months: this kit may be obsolete within a year.

What stays human is choosing the questions, where to point the tools, and judging what they return.

Rule 12 builds that in: you approve every protocol and every confirmed finding. Staying in the loop is part of the method, not a wish.

Use the agent to learn

Ask it to explain, then explain it back

Derive by hand, then let it check you

Read the digest, then the paper

Let a grilling skill question your plan

The wiki is written for you too: read it

A result nobody understands, remembers or communicates is wasted computation.

Start tomorrow: point your agent at the repo

```
Install research-knowledge-tree from github.com/bonneted/research-knowledge-tree,
make this folder an inquiry, and start it.
My question is: …
```

Or by hand

```
uv tool install \
  git+https://github.com/bonneted/research-knowledge-tree@v2.0.0
```

`rkt init` then: “start the inquiry”

You need

An agent: Claude Code, Codex, Gemini CLI…

Obsidian, to read the wiki

Zotero: recommended, not required

Start small: one inquiry, five papers, one question.

Damien Bonnet-Eymard

KU Leuven

Reliable & Robust Design (R2D)

iiw.kuleuven.be/onderzoek/R2D

Mecha(tro)nic System Dynamics

mech.kuleuven.be/en/mod

Flanders Make

flandersmake.be